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WORKSHEET 9

AIM: To design and implement a package-like structure using schema in PostgreSQL that contains procedures/functions and shared cursors to retrieve employee details, demonstrating modular and reusable database programming.

S/W Requirement: **Database Management System:**

- Oracle Database Express Edition (Oracle XE)
- PostgreSQL Database

Database Administration Tool / Client Tool:

- Oracle SQL Developer (for Oracle XE)
- pgAdmin (for PostgreSQL)

OBJECTIVES:

To implement modular database programming using schema (as package equivalent), functions, and shared cursors in PostgreSQL.

PROBLEM STATEMENT:

In enterprise systems, related database operations must be grouped together for better maintainability, reusability, and performance. While Oracle provides packages, PostgreSQL achieves similar functionality using schemas and functions.

PRACTICAL/EXPERIMENT STEPS:

Design a package-like structure that:

1. Contains procedures/functions to fetch employee details
2. Uses a shared cursor to retrieve employee records
3. Demonstrates execution of functions

TABLE STRUCTURE:

```
drop table employees;

CREATE TABLE employees (
    emp_id SERIAL PRIMARY KEY,
    emp_name VARCHAR(50),
    salary NUMERIC,
    dept_id INT
);
```

Data Output | Messages | Notifications

```
CREATE TABLE

Query returned successfully in 290 msec.
```

```
INSERT INTO employees (emp_name, salary, dept_id) VALUES
('Aman', 50000, 10),
('Riya', 60000, 20),
('Raj', 55000, 10),
('Neha', 70000, 30);
```

1. IMPLEMENTATION

Package Creation (Schema in PostgreSQL)

PROGRAM:

```
CREATE SCHEMA IF NOT EXISTS emp_package;

CREATE OR REPLACE FUNCTION emp_package.get_all_employees()
RETURNS refcursor AS $$
DECLARE
    emp_cursor refcursor := 'emp_cur';  -- Named cursor
BEGIN
    OPEN emp_cursor FOR
        SELECT * FROM employees;

    RETURN emp_cursor;
END;
$$ LANGUAGE plpgsql;
```

OUTPUT:

```
CREATE OR REPLACE FUNCTION emp_package.get_employee_by_dept(p_dept_id INT)
RETURNS refcursor AS $$
DECLARE
    emp_cursor refcursor := 'dept_cur';  -- Named cursor
BEGIN
    OPEN emp_cursor FOR
        SELECT * FROM employees
        WHERE dept_id = p_dept_id;

    RETURN emp_cursor;
END;
$$ LANGUAGE plpgsql;
```

```
BEGIN;

SELECT emp_package.get_all_employees();

FETCH ALL IN "emp_cur";

COMMIT;
```

	emp_id [PK] integer	emp_name character varying (50)	salary numeric	dept_id integer
1	1	Aman	50000	10
2	2	Riya	60000	20
3	3	Raj	55000	10
4	4	Neha	70000	30

```
BEGIN;

SELECT emp_package.get_employee_by_dept(10);

FETCH ALL IN "dept_cur";

COMMIT;
```

	emp_id [PK] integer	emp_name character varying (50)	salary numeric	dept_id integer
1	1	Aman	50000	10
2	3	Raj	55000	10

LEARNING OUTCOMES:

- Understand the concept of packages and their equivalent in PostgreSQL
- Differentiate between schema and package
- Implement functions with shared cursors

- Execute cursor-based data retrieval using FETCH
- Develop modular and reusable database logic used in organizations like Oracle Corporation and SAP

CONCLUSION:

The experiment successfully demonstrated the implementation of a package-like structure in PostgreSQL using schema and functions. By using shared cursors and modular logic, efficient and reusable database operations were achieved. This approach aligns with real-world enterprise practices, improving maintainability and scalability of database systems.